FRONT-END CAPSTONE SDLC FRAMEWORK

Proposed Aligned Industry-Standard Software Development Life Cycle

ACADEMIC–INDUSTRY ALIGNMENT

This framework refines the traditional 8-phase SDLC into an industry-aligned methodology specifically calibrated for front-end web application capstones. It incorporates modern engineering practices including **UI/UX Design Systems, Client-Side Architecture, WCAG 2.1 Accessibility, Automated CI/CD Deployment, and Product Monetization Governance**.

1. COMPARATIVE MAPPING: PROPOSED VS. INDUSTRY-STANDARD SDLC

Proposed Phase	Refined Industry-Standard Phase	Key Additions & Standard Terminology
1. Client Requirement	**Requirement Elicitation & UI Scope**	Define User Stories, Scope Boundaries, Non-Functional Requirements (NFRs), and MoSCoW feature prioritization.
2. Literature Survey	**UI/UX Benchmarking & Tech Stack**	Competitor UI audits, design system selection, CSS architecture planning, and client storage/fetch strategy.
3. Module Formulation	**UI Architecture & Component Design**	Figma wireframes & prototypes, CSS design tokens, Component Trees, and JSON API mock contracts.
4. Coding	**Front-End Dev & Data Integration**	Semantic HTML5, CSS Flexbox/Grid layouts, ES6+ JS logic, dynamic DOM updates, and REST API `fetch()` integration.
5. Testing	**Verification, Validation & a11y Audit**	Regex form validation tests, multi-device viewport breakpoint checks, cross-browser audits, and WCAG 2.1 accessibility.
6. Deployment	**Static Asset DevOps & Cloud Hosting**	Code minification, asset optimization, Git branching strategy, automated GitHub deployment to Vercel/Netlify with HTTPS.
7. Tech Transfer	**Style Guide & Technical Documentation**	Component library specs, Developer Onboarding `README.md`, API documentation, and Capstone Project Thesis.
8. Maintenance	**Core Web Vitals & UI Retrospective**	Console error logging, Lighthouse score monitoring (>90), beta user feedback loops, and future enhancement roadmap.

2. DETAILED PHASE EXECUTION & TECHNICAL ACTIVITIES

Elicitation & Scope — Phase 1

Goal: Define client requirements & project boundaries.

- Gather input/output constraints & user personas.
- Define NFRs: Target viewports, loading speed (<2s).
- **Deliverable:** SRS Document & Scope Matrix.

Benchmarking & Stack — Phase 2

Goal: Select tools & analyze domain patterns.

- Audit competitor UI layouts & micro-interactions.
- Select CSS framework/custom properties strategy.
- **Deliverable:** Tech Stack Justification Table.

UI Architecture & Design — Phase 3

Goal: Build visual blueprint before coding.

- Create Figma wireframes & interactive prototypes.
- Define CSS variables (colors, typography tokens).
- **Deliverable:** Figma Prototype & Component Tree.

Development & Integration — Phase 4

Goal: Implement responsive & interactive UI.

- Write semantic HTML5 & Flexbox/Grid CSS.
- Implement ES6+ JS, DOM events & `localStorage`.
- **Deliverable:** Clean Git Code Repository.

Verification & Accessibility — Phase 5

Goal: Ensure application stability & quality.

- Test Regex form validation & defensive edge cases.
- Audit WCAG 2.1 contrast ratios & keyboard focus.
- **Deliverable:** Test Matrix & Lighthouse Report.

DevOps & Cloud Deployment — Phase 6

Goal: Release application to production.

- Optimize image assets & minify CSS/JS bundles.
- Set up automated GitHub deployment to Vercel.
- **Deliverable:** Live HTTPS Production URL.

Documentation & Hand-off — Phase 7

Goal: Package deliverables for stakeholders.

- Write technical Developer Setup `README.md`.
- Document UI Style Guide & API Endpoint Specs.
- **Deliverable:** Final Project Report & User Guide.

Monitoring & Retrospective — Phase 8

Goal: Evaluate launch health & future scope.

- Collect beta user feedback on UI intuitiveness.
- Log client-side console errors & map v2 features.
- **Deliverable:** Post-Mortem & Feature Roadmap.